Two-mode Network Analysis of Culture Concepts

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Abstract

Kroeber and Kluckhohn's classic 1952 monograph *Culture: A Critical Review of Concepts and Definitions* (Kroeber and Kluckhohn, 1952) brought together over a hundred scholarly definitions of culture and indexed the key concepts they discuss. We build one network from that index to explore how concepts and definitions are connected. In the *definitions network*, concepts like **society** and **beliefs** are linked to 171 definition codes, grouped into seven categories and credited to the author or authors that K&K index under each. Each code is one quotation, identified by its author, year, page numbers and opening words. We present basic statistics about the network, visualize the data using correspondence analysis and spanning trees, and highlight the key concepts and definitions that anchor it. Data-driven types found in the concept profiles and in the wording of the definitions do not recover K&K's categories at better than chance.

1 Data and networks

We build one network from the Index B of Kroeber and Kluckhohn (1952), hereafter K&K. The *definitions network* links 470 concepts—concrete ideas like **society**, **customs**, and **beliefs**—to 171 definitions of culture, which K&K number as codes. Each code is credited to the author or authors K&K index under it, and falls into one of seven categories, which K&K label with the capital letters A through G: A Descriptive, B Historical, C Normative, D Psychological, E Structural, F Genetic, and G Incomplete.[1] Of the 171 codes, 169 appear in K&K's own author index. The other two, C-I-18a and D-II-16, are attributions cited in the main text of the book that the index does not list. Boroch (2016) quotes two parts of four K&K codes separately, lettered (a) and (b): A-3, E-5, F-II-4 and F-II-7. We keep each of these as two codes, A-3(a) and A-3(b) and so on, which is why the eight variant codes replace four. K&K print one index entry per code, so each variant starts with its parent's concept links. For 29 of the 41 concept links copied in this way, we kept the link only on the variant whose quotation contains the concept's name; the other 12 remain on both variants. That assignment is our reading of the quotation text, not taken from anything directly printed in the K&K index. Concepts have consistent names; where a concept appears under different headings in K&K's source text, we treat it as a single concept. Each code is one quotation, identified by its author, year, page numbers and opening words. Section A of the appendix describes how each code was matched to its quotation and how the a and b variants were built.

The 171 × 470 definitions network naturally breaks into 10 separate components, with the largest containing 626 nodes. Our analysis focuses on this largest component, which contains virtually all the meaningful connections. We use maximum spanning trees across multiple projections to identify the strongest connections in different views of the network.

[1] K&K further divide three of the seven categories into numbered subcategories: Normative (C) into C-I and C-II, Psychological (D) into D-I through D-IV, and Genetic (F) into F-I through F-IV. The other four categories have no subcategories. Codes that sit in a subcategory carry its numeral, as in C-I-18a or F-II-4 below.

2 Overview of the network

Table 1 gives summary statistics for the definitions network. The first three rows show node and edge counts. The "degree" rows count connections of each node type; a concept's degree is how many definitions mention it, while a definition's degree is how many concepts it contains. Concepts and definitions differ substantially in their average degree.

Table 1: Summary statistics for the definitions network. Statistics describe the full network, before the restriction to the largest component's 2-core used in the analyses. Average nearest-neighbor degree (k_{nn}) is the average degree of a node's neighbors; degree correlation shows how a node's degree relates to the degree of its neighbors. Degree variances are sample variances (divisor $n-1$) over all nodes of each type, including isolates.

	Network
Edges	1201
Density	0.015
Degree correlation r	−0.076
Concepts ($N = 470$)	
Mean degree	2.56
Minimum degree	1
Maximum degree	36
Degree variance	16.34
Mean k_{nn}	10.18
Definitions ($N = 171$ codes, 5 isolated)	
Mean degree	7.02
Minimum degree	0
Maximum degree	25
Degree variance	20.52
Mean k_{nn}	9.22

The degree correlation is −0.076: concepts with many connections tend to link to definitions with few, and vice versa, though the association is weak. The nearest-neighbor rows show the other side of this pattern. A concept's neighboring definitions contain 10.2 concepts on average, and a definition's neighboring concepts appear in 9.2 definitions on average, so concepts attach preferentially to heavily used definitions and definitions attach to widely shared concepts.

Figure 1 shows the degree distributions for the concept nodes (left) and the definition nodes (right). Most concepts are lightly connected: 79% appear in two or fewer definitions, while a small minority of concepts, such as **social**, **society**, **group**, **behavior** or **ideas**, appear in many definitions (see Table 2). This leads to the classic "long tailed" degree distribution for definitions observed in the left-hand side of Figure 1.[2]

Definitions are more evenly connected, with the average definition covering 7 concepts (median

[2]Over the full range of concept degrees, a discrete power law fits better than a truncated exponential (Vuong $z = 6.1$, $p < 0.001$; $\hat{\alpha} = 2.13$), suggesting a mechanism of "preferential attachment" of definitions to already popular concepts (Barabási and Albert, 1999). The advantage shrinks for degrees of 2 and up ($z = 1.9$, $p = 0.06$) and vanishes for degrees of 3 and up ($|z| < 1$), so the tail alone does not distinguish the two.

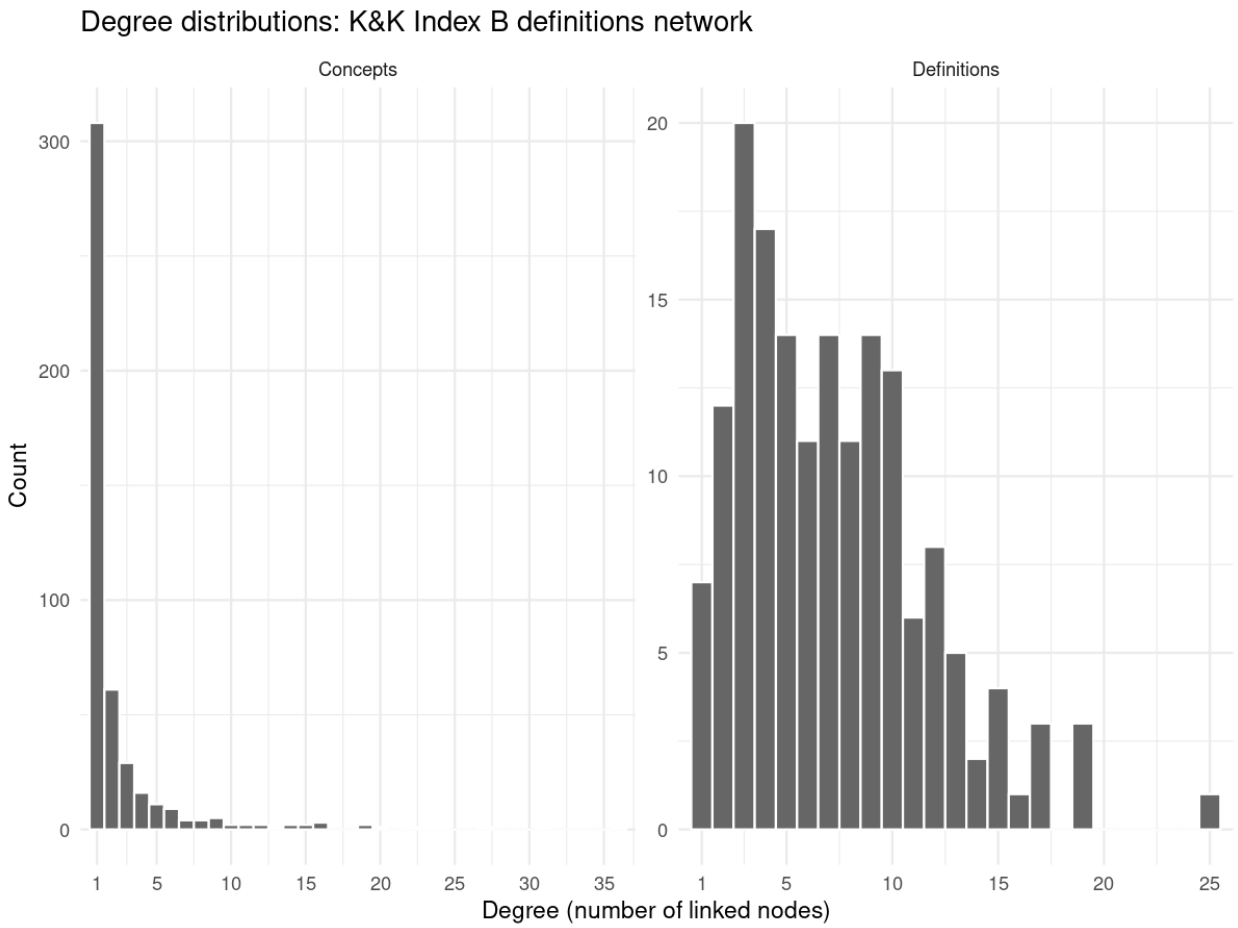


Figure 1: Degree distributions of the definitions network. Left: sub-element (concept) nodes. Right: definition (code) nodes.

7) and a few covering many more, up to 25.[3] The degree distribution differs from that of the definitions accordingly, coming closer to a bell-shape with a well-behaved (representative) mean. Thus, few definitions are "lean" (relying on a few concepts) and even fewer are conceptually profligate, with most definitions relying on a moderate number of concepts. This means that while most definitions help themselves to a small set of ubiquitous concepts, they also draw from a wider set of concepts not necessarily shared across definitions.

3 Maximum Spanning Trees

By construction, the two-mode definitions network has no ties between two concepts or between two definitions, so we analyze the implied concept-to-concept ties and the definitions-to-definitions connections via multiple one-mode projections (Everett and Borgatti, 2013). Following Breiger (1974), two concepts are tied if they share at least one definition, with weight equal to the number

[3] Across the 171 codes, the number of words in each quotation correlates with the number of concepts the code links to: Pearson $r = 0.66$, Spearman $\rho = 0.64$. Excluding the five codes with no concept links gives $r = 0.66$ and $\rho = 0.63$ on 166 codes. Word counts use the English translation for the four non-English quotations and exclude Boroch's ellipsis markers.

Table 2: Highest-degree nodes of the definitions network. Concept degree counts the definitions that contain the concept; definition degree counts the concepts the definition contains. The last two columns rank definitions by degree per 10 words of quotation, using the English translation where one exists. Per-word rankings favor short quotations: across the 171 codes the per-word value correlates −0.33 with word count. Degrees come from the full verified network. Ties at the cutoff are included.

Concepts, by degree		Definitions, by degree		Definitions, by degree per 10 words	
Concept	Degree	Definition (code)	Degree	Definition (code)	Per 10 words
SOCIAL	36	F-I-8 (Reuter, 1939)	25	F-I-15 (Carr, 1945)	5.56
SOCIETY	31	C-I-6 (Gillin and Gillin, 1942)	19	A-3(a) (Dixon, 1928)	5.00
GROUP	27	D-I-6 (Lundberg, 1939)	19	D-III-2 (Young, 1934)	4.29
BEHAVIOR	24	F-II-8 (Taylor, 1948)	19	D-II-4 (Kluckhohn, 1942)	4.29
IDEAS	23	A-8 (Hiller, 1933)	17	F-III-1 (Bain, 1942)	4.29
TOTAL	22	C-I-17 (Frank, 1948)	17	D-II-12 (Wilson and Kolb, 1949)	4.23
SUM	20	D-I-7 (Panunzio, 1939)	17	E-7 (Gillin, 1948)	4.21
BELIEFS	19	F-I-3 (Folsom, 1931)	16	A-3(b) (Dixon, 1928)	4.17
INDIVIDUALS	19	A-15 (Kluckhohn and Kelly, 1945)	15	B-13 (Groves and Moore, 1940)	4.17
CUSTOMS	16	E-3 (Ogburn and Nimkoff, 1940)	15	A-1 (Tylor, 1871)	4.14
HABITS	16	F-I-19 (Murdock, 1949)	15	A-2 (Wissler, 1920)	4.12
LEARNING	16	F-III-4 (White, 1949)	15	C-II-2 (Thomas, 1937)	4.09

they share. Likewise, two definitions are tied through the concepts they share. In matrix form, the projections are:

$$W_{\text{concept}} = AA^{\top}, \qquad W_{\text{def}} = A^{\top}A, \tag{1}$$

Where $(W_{\text{concept}})_{ij} = \sum_k a_{ik}a_{jk}$ counts the definitions that include both concepts i and j. Each projection is a weighted graph whose edges are the off-diagonal pairs with positive weight. We build projections from the 2-core of the network's largest component.

Projected graphs are typically dense and fully connected making it hard to detect their underlyinng structure. To do so, we extract the "backbone" (Neal, 2014) of each projection by extracting the maximum spanning tree (MST). The MST of each projection is found by computing the minimum spanning tree on negated weights, $-w$. Kruskal's (1956) method sorts edges from largest weight to smallest and adds each unless it would close a cycle, producing a tree that maximizes $\sum_{e\in T} w_e$. Because the 2-core is connected, each tree has exactly $|V| - 1$ edges. In the visualizations, node size shows betweenness within the tree and node color shows the community structure detected using Newman's (2006) leading-eigenvector method applied to the tree.

The concept-level tree (Figure 2) shows how 160 concepts cluster in the concept projection and the definition-level tree (Figure 3) shows how 149 definitions cluster in the definitions projection. Table 3 lists the top 10 concepts and definitions by betweenness in their maximum spanning trees.

In the concept tree (Figure 2), the six largest nodes—**group**, **customs**, **social**, **behavior**, **society**, and **beliefs**—are the top six of Table 3. **group** sits at the center of a pink region of **communication** and **organization** vocabulary (**communication**, **organization**, **folkways**, **systems**, **designs for living**). **social** heads a green region of social-organizational vocabulary (**social heritage**, **institutions**, **social needs**, **patterns**, **goods**), and **customs** heads an olive region of tradition and adjustment (**social group**, **accumulated treasury**, **adjust**, **association between persons**). **behavior** and **society** head the blue and magenta regions on the left, which hold the historical and learned-behavior vocabulary (**historically**, **standards**, **learned behavior**,

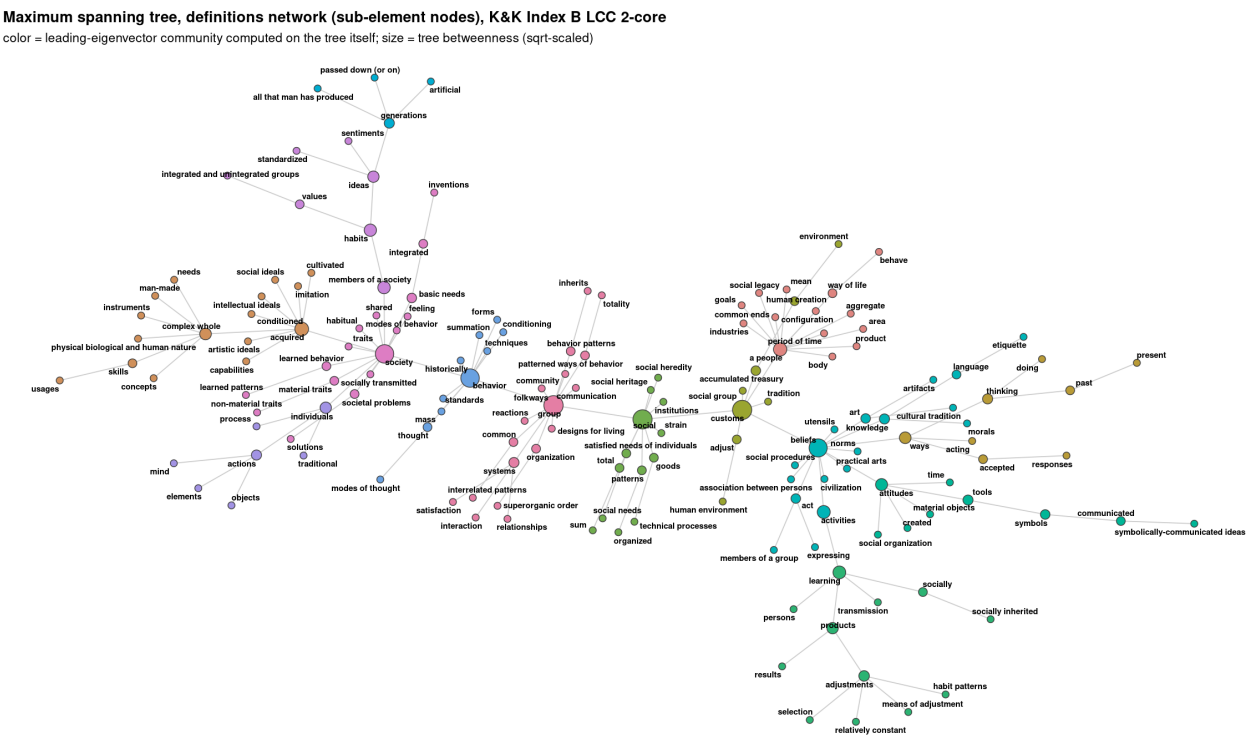


Figure 2: Maximum spanning tree of the definitions network, sub-element (concept) nodes. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

socially transmitted). **beliefs** heads the teal region of **norms**, **knowledge** and **artifacts** on the right, with **norms**, **practical arts**, **act** and **civilization**. A peripheral branch below **beliefs** runs through **activities** and **learning** to **products** and **adjustments**. This chain collects the acquisition and **transmission** vocabulary of K&K's "Genetic" definitions and hangs off the core through a single path, so these **ideas** connect to the rest of the network only transitively. At the top left, **ideas**, **habits**, **values** and **generations** form a second branch that joins **society** through **habits**.

Two nodes at the edges of this picture need separate mention. **a people**—an alternative reference to a group or collective—is the link between the olive **customs** region and a branch on the right that includes **way of life** and **period of time**. Removing it disconnects that branch from **customs**. Its strongest ties are to **way of life** (six shared definitions) and **customs** (five), and its fifteen definitions come from all seven groups, so it is a broadly shared concept rather than one tied to a single group. This speaks to the penchant to define culture as the customs or way of life common to "a people." **complex whole**, the classic locution due to the "original" definition by Tylor (1871), heads an orange cluster on the far left. Its tree neighbors are **concepts**, **instruments**, **man-made**, **needs**, **physical biological and human nature** and **skills**, together with **acquired**, which it shares with three definitions. The cluster reaches **society** only through **acquired**, and its six definitions come from Groups A (Descriptive), B (Historical) and D (Psychological).

The definition tree (Figure 3) puts its three largest nodes at the center of the tree—the Gillin and Gillin (1942) quotation (C-I-6), the Murdock (1949) quotation (F-I-19), and the Bidney (1947) quotation (A-16)—with Gillin and Gillin the largest of all. Gillin and Gillin (1942) heads the largest community (24 definitions, drawn from all seven groups), and Bidney (1947) sits in the same community. We have already seen that Gillin and Gillin (1942) is one of the most central

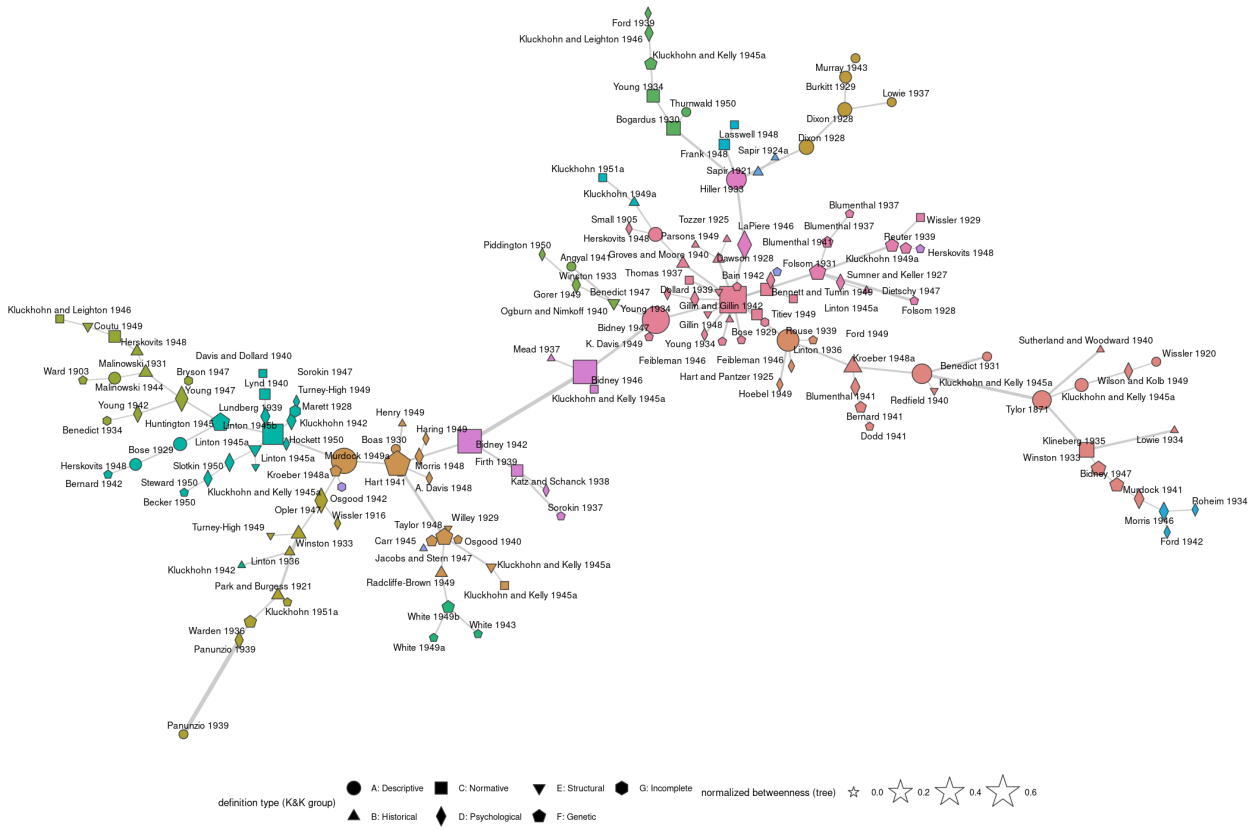


Figure 3: Maximum spanning tree of the definitions network, code (definition) nodes labeled by the author and year of their quotation. Shape is the K&K definition group (A–G, Descriptive to Incomplete); color is the leading-eigenvector community on the tree; size is betweenness in the tree.

definitions by degree (using lots of concepts). It is also wordy (64 words), but not the wordiest (it sits at the 90th percentile of wordiness). So its bridging role is not a simple function of its length, but of the concepts it uses, as there are way wordier definitions that are not as central. Here's the definition in full, with the concepts it uses in bold:

> The **customs**, **traditions**, **attitudes**, **ideas**, and **symbols** which **govern social behavior** show a wide variety. Each **group**, each **society** has a set of **behavior patterns** (overt and covert) which are more or less **common** to the members, which are passed down from **generation** to **generation**, and **taught** to the **children**, and which are constantly liable to **change**. These **common** patterns we call the culture...

C-I-6 is the most central definition in the tree, and its position is not a simple function of its length. At 64 words it sits at the 90th percentile of quotation length, well short of the longest quotation (142 words), yet it leads the ranking by both degree and betweenness. Nor is its position a function of its vocabulary alone. Its concepts include five of the six most central concepts of the concept tree—**group**, **social**, **behavior**, **society**, and **customs**—and the mean betweenness of its incident concepts sits at the 87th percentile of all codes. But a central vocabulary is not sufficient: 54 other codes share C-I-6's two most central concepts, and their mean betweenness is

0.05 against C-I-6's 0.71. What is distinctive about C-I-6 is where it sits in the definition tree: it bridges definitions, and it does so with a concept repertoire that many other definitions share.

Bidney's (1947) is in many ways more impressive. It is the third most central definition in the tree, but it is short (only 34 words). Its links are a mix: it is tied to three of the six most central concepts (**social**, **behavior**, and **society**) alongside six concepts with no tree betweenness at all, and the mean betweenness of its linked concepts sits at the 68th percentile. Its position in the tree is what makes it central: removing it splits the tree into four components, including its two largest regions. Here is the definition in full. Of its linked concepts, only *acquired* and *society* appear verbatim in the text; **behavior** is linked but does not appear in the quotation:

> Culture is that complex whole which includes artifacts, beliefs, art, all the other habits **acquired** by man as a member of **society** and all products of human activity as determined by these habits.

Two of the other Bidney definitions lead us gradually from the right side of the tree to the left side. First, we have C-II-4 (Bidney, 1946), the fourth most central definition in the tree. It is a "Normative" definition according to K&K, and it shares its community with C-II-3 (Bidney, 1942) but not with C-I-6 (Gillin and Gillin, 1942) or A-16 (Bidney, 1947). Its quotation is 40 words long, and it links to 13 concepts, three of which are among the four most central concepts of the concept tree: **social**, **behavior**, and **society**. Its quotation uses "social" only inside "social ideals," so **social** is not bolded there:

> An integral or holistic concept of culture comprises the **acquired** or **cultivated behavior**, **feeling**, and **thought** of **individuals** within a **society** as well as the **patterns** or **forms** of intellectual, social, and artistic ideals which human societies have professed **historically**.

Finally, we have C-II-3 (Bidney, 1942), the fifth most central definition in the tree. It is also a "Normative" definition according to K&K, and it shares its community with C-II-4 but not with C-I-6 or A-16. Its quotation is 40 words long, and it links to 11 concepts, two of which are among the four most central concepts of the concept tree (**behavior** and **society**), along with **members of a society**, the seventh most central concept:

> A culture consists of the **acquired** or **cultivated behavior** and **thought** of **individuals** within a **society**, as well as of the intellectual, artistic, and social ideals which the members of the society profess and to which they strive to conform.

Murdock (1949) (F-I-19) and the Kroeber (1948) quotation A-17 are tree neighbors and share a second community at the left of the center. Murdock's is the second most central definition in the tree. Its 70 words link to 15 concepts, three of which are among the four most central concepts of the tree (**group**, **behavior**, and **society**), and it has eight tree neighbors, including C-II-3 and A-17. Its quotation frames culture as the product of a process, the "interaction of learning and society":

> The **interaction** of **learning** and **society** thus produces in every human **group** a **body** of **socially transmitted adaptive behavior** which appears super–individual because it is **shared**, because it is **perpetuated** beyond the individual life span, and because its quantity and quality so vastly exceeds the capacity of any single person to achieve by his own unaided effort. The term culture is applied to such **systems** of **acquired** and transmitted **behavior**.

Kroeber's A-17 (65 words; pp. 8–9) is sixth by tree betweenness and links to 13 concepts. Only one of them, **behavior**, is among the most central concepts of the concept tree. The others include **learning** (0.145), **habits** (0.133), and **ideas** (0.062), while **mass**, **techniques**, **totality**, and **transmission** have no tree betweenness at all. Its tree neighbors are C-I-8, D-II-8, F-I-19, and F-II-6. Its quotation lists the components of culture and then frames it as a total product:

> [. . .] the **mass** of learned and transmitted **motor reactions**, **habits**, **techniques**, **ideas**, and **values** – and the **behavior** they induce – is what constitutes culture. Culture is the special and exclusive product of men, and is their distinctive quality in the cosmos [. . .]. Culture [. . .] is at one and the same time the **totality** of products of social men, and a tremendous **force** affecting all human beings, **socially** and **individually**.

This is consistent with the table, where five of the top ten definitions carry a C (Normative) or F (Genetic) code and the others are Descriptive, Historical or Psychological. Degree and betweenness dissociate in this ranking: C-I-6 has both the highest betweenness and the highest tree degree (17), acting as the main hub of its community, while F-I-19 has the second-highest betweenness with only eight ties, sitting on the paths that join its cluster to the rest of the tree. One author supplies a striking share of the bridges, as Bidney holds three of the ten most central definitions (A-16, C-II-4, and C-II-3); Kroeber (1948) is the only other work with two codes in the top ten (A-17 and B-18), and Linton holds two more, C-I-8 (Linton 1945b, a chapter) and A-10 (Linton 1936, a book). Away from the center, the Descriptive definitions—the two Dixon 1928 quotations (A-3(a) and A-3(b)), Burkitt 1929 (A-5), Lowie 1937 (A-10a), and Murray 1943 (A-12)—form a separate olive branch in the upper right, and the Descriptive definitions Tylor 1871 (A-1) and the Kluckhohn and Kelly 1945 chapter (A-14) run along the right side, showing that the earliest and simplest definitions connect to the core through fewer, longer paths.

The two trees' betweenness rankings are each their own structural fact rather than echoes of one another. A concept's tree betweenness barely predicts the betweenness of the definitions that contain it ($r = 0.08$), and the reverse holds as well ($r = 0.04$). Concepts gain tree betweenness by bridging concept regions, not by riding on central definitions: **group** owes its top rank to its position among concepts, not to its definition partners. The converse also appears. C-I-8 and A-10 sit high in the code tree while drawing mostly on concepts in the bottom half of the concept betweenness ranking, so a definition can bridge definitions without containing any bridge concept. C-I-6 shows the same from the other side: its central vocabulary is shared by dozens of unremarkable codes, so its own betweenness reflects its position in the definition tree rather than the concepts it uses. The apparent association between the top concepts and the top code is largely one hub: the leading concepts all connect to C-I-6, so its 0.71 betweenness dominates their maximum incident value.

The works themselves barely cluster. The 149 codes of the tree carry 120 distinct author–year labels—a repeated label is a second quotation from the same work—and the 19 labels that appear two or more times almost never stay together: 15 of them spread over two or more communities. Kluckhohn and Kelly (1945)'s chapter, the most quoted work in the corpus with seven of its eight codes in the tree, reaches four communities scattered across the tree, and Herskovits (1948) likewise spans four. The three a/b quotation pairs do sit together, one pair per community (Dixon, 1928; Blumenthal, 1937; Feibleman, 1946), as those variants share most of their concept links; Linton (1945a) is the partial exception, with its two E-5 variants together and its third code (C-I-9) in another community. The regions of the tree organize concept vocabulary, not the works the definitions were drawn from.

The six most central concepts, **group**, **customs**, **social**, **behavior**, **society** and **beliefs**, are the top of Table 3, and each anchors one of the color regions of the concept tree. The most central

Table 3: Top 10 nodes by Freeman-normalized betweenness in the definitions network MST.

Node type	Node	Author-year	Norm. Betw. (%)	Degree
Concepts	GROUP		59.39	12
	CUSTOMS		58.04	7
	SOCIAL		57.38	10
	BEHAVIOR		50.83	10
	SOCIETY		50.66	14
	BELIEFS		48.40	13
	ACQUIRED		17.81	9
	A PEOPLE		15.72	13
	ACTIVITIES		14.04	2
	LEARNING		13.22	5
Definitions	C-I-6	Gillin and Gillin 1942	71.45	17
	F-I-19	Murdock 1949	55.57	8
	A-16	Bidney 1947	53.42	4
	C-II-4	Bidney 1946	50.94	4
	C-II-3	Bidney 1942	50.88	3
	A-17	Kroeber 1948	43.88	4
	C-I-8	Linton 1945b	32.15	6
	A-10	Linton 1936	27.01	5
	B-18	Kroeber 1948	22.97	3
	D-II-5	LaPiere 1946	19.42	2

Note: Definition codes use the format X-Y-Z, where X is a letter (A–G) for K&K's seven definition categories (Descriptive, Historical, Normative, Psychological, Structural, Genetic, Incomplete), Y is a roman numeral for the subcategory, and Z is the number of the definition within it. Variants such as A-3(a) and A-3(b) are the two quotations of one K&K code, and they are counted as separate codes (see the data section). The author–year column names the source of each definition's quotation, as Boroch attributes it; the same author and year label the node in Figure 3.

definition is C-I-6, at 71% of the maximum possible, a single Normative code. In the concept tree the acquisition and **transmission** vocabulary sits on peripheral branches rather than at the core, suggesting that Genetic concerns, however common in the definitions themselves, connect to the rest of the conceptual structure only through a narrow path.

4 Data-driven types: non-negative matrix factorization

The spanning trees and the community colors above classify nodes hard: each code belongs to exactly one community of the tree. Non-negative matrix factorization (NMF) (Lee and Seung, 2001) offers a soft, parts-based alternative. It approximates a non-negative matrix V by a product WH with all entries non-negative, so every code carries a loading on each factor and every factor is an additive mixture of the original features, with no subtractive terms. We ask how far data-driven types agree with K&K's own seven categories A–G (Descriptive to Incomplete), first from the concept profiles of the definitions—which concepts each definition uses—and then from the wording of the definitions. Both factorizations use the same base-R implementation throughout (Euclidean multiplicative updates, deterministic given a seed, with ten random restarts per value of k), and both fix k at seven, one factor per K&K group. Fixing k this way is a choice, not a result: the reconstruction error falls monotonically in k without an elbow, and restart agreement is irregular (mean adjusted Rand index 0.51 to 0.67 over $k = 3$ to 8 for the concept profiles, 0.48

to 0.70 for the wording), so neither criterion selects a number of factors on its own.

4.1 Types from the concept profiles of the definitions

The first factorization asks the question of the network itself. Its matrix is the definitions network's own biadjacency on the largest component's 2-core: 149 definition codes, each described by the 160 concepts it uses, coded 1 when the definition contains the concept and 0 otherwise. The factorization is run at $k = 7$, one factor per K&K group.

The factors read as recognizable concept profiles rather than K&K's categories (Figure 4), and we label them by their leading concepts: *social organization* (**social**, **social heritage**, **institutions**, **patterns**), *group and way of life* (**group**, **behavior**, **common**, **a people**), *norms and knowledge* (**beliefs**, **customs**, **ways**, **knowledge**), *society and acquired behavior* (**society**, **acquired**, **members of a society**, **individuals**), *transmitted ideas* (**ideas**, **habits**, **generations**, **attitudes**), *complex whole* (**total**, **sum**, **tools**, **activities**), and *systems and skills* (**systems**, **organization**, **satisfaction**, **skills**). The last factor is diffuse: its leading loadings are an order of magnitude smaller than the others', so its label is the least secure. Against the A–G groups (Descriptive to Incomplete) the agreement is indistinguishable from chance (adjusted Rand index 0.002, against a permutation 95th percentile of 0.016), and no factor puts more than 40% of its codes in one group. The subgroup-level view is in the appendix (Figure 6).

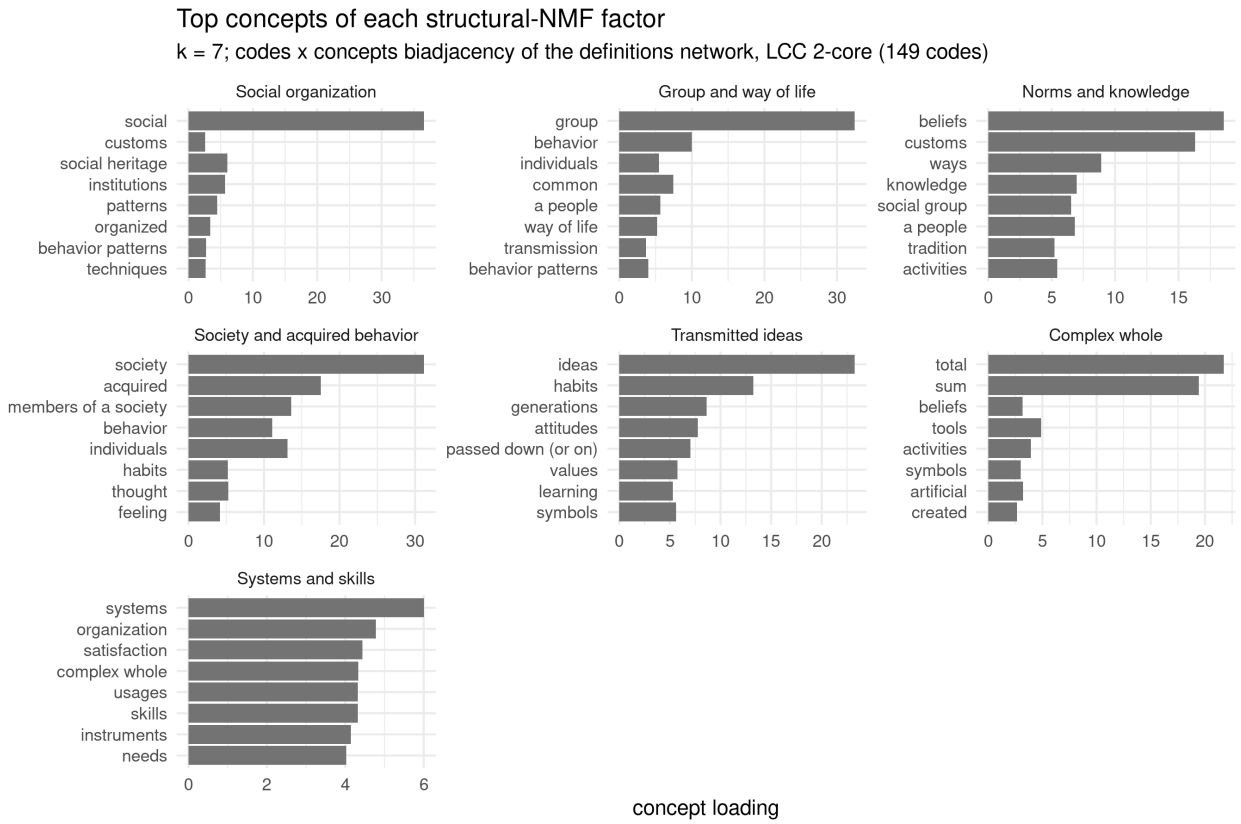


Figure 4: Top concepts of each structural-NMF factor at $k = 7$, from the codes × concepts biadjacency of the definitions network's largest-component 2-core (149 codes).

4.2 Types from the wording of the definitions

The second factorization asks the question of the definition texts themselves, independently of both K&K's coding and the concepts the definitions use.

The second factorization asks the question of the definition texts themselves, independently of both K&K's coding and the concepts the definitions use. The corpus holds 171 definition codes carrying a quotation: the texts from Boroch's Table 2 mapped to K&K codes through the crosswalk, the appendix-only texts, and one statement K&K prints itself (a Part II page footnote). Three Table 2 columns that repeat another column's quotation word for word are not counted again. Documents average about thirty-four words (median 28), so the term–document matrix is built restrictively: tokens of three letters or more, a stopword list that removes the words nearly every definition contains (culture, society, social, man, and the general definitional vocabulary total, whole, learned, acquired), and terms kept to a document frequency between two documents and half the corpus, leaving 403 terms weighted by TF–IDF. The four quotations K&K and Boroch print in French or German enter as English translations (K&K's own rendering for Maquet, ours for the other three, not yet checked against the printed pages). The factorization is run at $k = 7$, one factor per K&K group.

The factors again find recognizable themes rather than K&K's categories (Figure 5), and we label them by their leading terms: *species-specific ability* (events, species, use, dependent, ability), *order and operation* (order, operation, self, superorganic, complex), *mental and holistic* (concept, mean, holistic, traits, mental), *goals and persons* (certain, rather, others, persons, goals), *symbolic process and adjustment* (concrete, process, symbolic, experience, adjust), *material and transmitted behavior* (material, non, transmitted, socially, institutions), and *passed down generations* (generation, passed, down, everything, common). On a corpus this small a factor can be driven by one author's phrasing, so the labels should be read as descriptive of the top terms, not as clean concepts. The labels for mental and holistic and goals and persons are the least clear, since their leading terms include generic words (rather, certain, some, mean). Against the A–G groups (Descriptive to Incomplete) the agreement is close to chance (adjusted Rand index 0.012, against a permutation 95th percentile of 0.016), and no factor has more than 54% of its codes in one group.

4.3 Interpretation

Neither the concept profiles nor the wording of the definitions recovers K&K's taxonomy. The two factorizations fail on different matrices—the binary concept codings of the network and the texts of the quotations—so the null result does not hinge on one representation. Two readings are possible. One is that K&K's groups classify the emphases of individual *statements*, and that these emphases cannot be recovered from a binary concept coding, or from quotations of about thirty words each. The other, which we find more plausible, is that the seven groups are editorial categories—useful for organizing a bibliography—rather than dimensions along which the corpus varies. This is consistent with the spanning-tree picture above, where the vocabulary the Genetic group concerns itself with sits on a peripheral chain of the concept tree rather than forming a region of its own, and with the community structure of the trees, which follows concept vocabulary, not authorial schools. The null result should be read against the method's limits as well: the number of factors was fixed to match the taxonomy under test rather than selected by the data, the short corpus makes TF–IDF factors fragile (restart agreement ranges from 0.48 to 0.70 for k from 3 to 8), and the quotations are the passages Boroch selected, not the authors' full works.

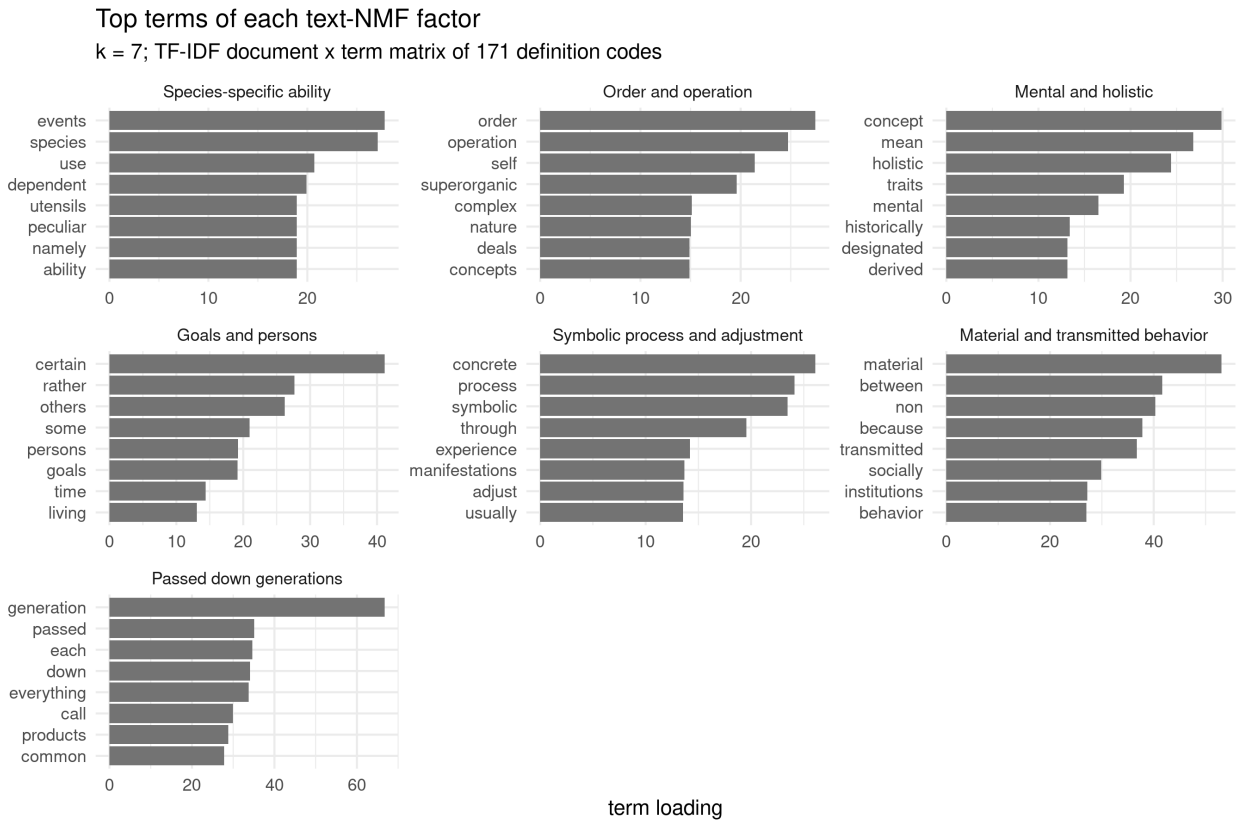


Figure 5: Top terms of each factor of the text NMF at $k = 7$, from the TF–IDF matrix of the 171 definition codes.

5 Limitations

The networks depend on the K&K Index B transcription and on the verified-code filter. Element names are not unique across headings, so same-named elements are merged, which can join unrelated entries. Several readings depend on OCR corrections recorded in the repository's manual-fix files; the validators check structure and round-trips, not the content of each edge. Two codes that are not in Index A are included: C-I-18a and D-II-16, from the main text. K&K's Part III statements (C-5, D-5, D-7, E-17) are excluded. The basis for each is documented in the repository. The four split codes and the 29 link decisions that separate their variants (see the data section) are judgments: the variants share one K&K index entry, and the text-based assignment of concept links is a reading that a reader can check against the quotations, not a printed fact. The definition-level trees and the text factors depend on these decisions.

A Constructing the code-level data

The analyses use one object: a table with a row for each of the 171 K&K definition codes. Each row gives the code's group and subgroup, its sub-elements, and the citation that identifies its quotation. This section records how that table was built, the choices made along the way, and the basis for each. Two sources are joined. K&K's Index A and Index B, read from a scan of the 1952 monograph, give each code its author, year and sub-elements. Boroch's Boroch (2016) gives each code a quotation with page numbers. Every join follows a stated rule, and each judgment is recorded with its basis in the project's provenance log. The analyses use one object: a table with a row for each of the 171 K&K definition codes. Each row gives the code's group and subgroup, its sub-elements, and the citation that identifies its quotation. This section records how that table was built, the choices made along the way, and the basis for each. Two sources are joined. K&K's Index A and Index B, read from a scan of the 1952 monograph, give each code its author, year and sub-elements. Boroch's (2016) article gives each code a quotation with page numbers. Every join follows a stated rule, and each judgment is recorded with its basis in the project's provenance log.

A.1 Sources and the unit

The unit of analysis is the code K&K attached to each definition, which corresponds to one printed index entry in K&K, and one quotation in Boroch (2016). Boroch's appendix has 171 codes. Boroch's Table 2 has 135 columns, 132 of which match an appendix code. The two sources number codes differently. Boroch renumbers within each group, so his A10 (Winston, 1933) is K&K's A-9. Codes are therefore matched by author, year, group and subgroup. Positional agreement is kept only as a check: of 100 matches made by name and year, 48 agree with the position.

A.2 Index A and Index B

Index A (authors) and Index B (sub-elements) were read from the OCR text layer and corrected by explicit rules. The corrections are recorded in a manual-fix table of 142 rows, each with its printed-page evidence. A further 10 rows recover codes that the OCR glued into element names, and 14 rows record name decisions. The project avoids general character rules, because a rule such as reading a y as a 5 silently produced codes from unrelated tokens. Each manual fix is keyed to a parsed token, and the checks confirm that every key matches one.

The universe is 171 codes. Of these, 169 appear in K&K's author index. The other two, C-I-18a (Maquet 1949) and D-II-16 (Aberle et al. 1950), are cited in main-text footnotes but not listed in the index; their Index B entries are read from the sub-element entries that carry them. Four Part III statement codes (C-5, D-5, D-7 and E-17) are excluded, since they are K&K's statements rather than Part II definitions.

A.3 Matching Boroch's codes to K&K's

The crosswalk matches each column of Boroch's Table 2 to a K&K code. Of the 135 columns, 117 match automatically when exactly one code fits, and 15 more are settled by hand decisions, each with a basis. Three of Boroch's Table 2 columns have no appendix entry of their own and carry no K&K code: A10a, C1-5a and F4-6. Each is a verbatim copy of another quotation, which leaves 132 columns with a K&K code. A10a repeats A10 (Winston, 1933), C1-5a repeats C1-5 (Firth, 1939), and F4-6 repeats F4-5, which is the entry Boroch's numbering misprints as F4-6. None of the three adds a quotation, so the object does not use them. The F4-5 quotation is Kluckhohn and

Kelly 1945 (printed p. 87), and it gives K&K's F-IV-5. K&K's printed codes A-10a (Lowie, 1937) and C-I-5a (Lynd, 1940) take their quotations from the appendix entries A12 and C1-6.

A.4 Quotations and citation keys

Each code has one quotation. Boroch supplies 170 of them: the 132 codes from Boroch's Table 2 and the 39 appendix-only codes, less the Sears addendum (C1-22), which is linked to a specialty code rather than to a network code. D-III-4 has no Boroch entry, and its quotation is K&K's printed footnote 14 (Roberts 1951, pp. 3 and 6, printed p. 60). Four quotations are not in English and are rendered from a translation table: B-17 and C-I-18a from French, and F-I-5 and F-II-3 from German. C-I-18a uses K&K's own printed English. The other three are working translations that have not been checked on the page images.

Each code is identified by a citation key: Boroch's author, year and page numbers, together with the first six words of the quotation. The key is unique for all 171 codes, with no missing values. The page numbers are Boroch's as printed and have not been checked against the source works. The key therefore identifies a quotation as Boroch reports it, and the page numbers are the first thing to check against a source.

A.5 Splitting the a and b quotations

Boroch quotes parts of four K&K codes separately, labeled (a) and (b). K&K prints one index entry for each of these codes, so the object had one code where Boroch has two quotations. On 2026-10-30 the project split each of the four codes into two variants (Table 4). The split follows Boroch's quotations and is not a distinction K&K prints. Each variant takes its own quotation and its own citation key.

Table 4: The four split codes and their variants.

K&K code	Variant	Boroch code and source	Page numbers
A-3	A-3(a)	A3, Dixon 1928	3
A-3	A-3(b)	A4, Dixon 1928	3
E-5	E-5(a)	E5, Linton 1945a (footnote 5)	5
E-5	E-5(b)	E6, Linton 1945a	32
F-II-4	F-II-4(a)	F2-4, Blumenthal 1937 (footnote 4)	3
F-II-4	F-II-4(b)	F2-5, Blumenthal 1937	12
F-II-7	F-II-7(a)	F2-8, Feibleman 1946 (tentative; footnote 7)	75
F-II-7	F-II-7(b)	F2-9, Feibleman 1946 (final)	76

Because K&K gives one index entry per code, each variant starts with a copy of its parent's sub-element links, 41 in all. A lexical screen then assigns 29 of the 41 to one variant. A link stays on the variant whose quotation contains the sub-element's name and not on the other. The 12 links where both or neither quotation contains the name remain on both variants. The screen is a word match, so a name used in a generic sense, such as "total" or "past," can be assigned wrongly. Each assignment is a reading of the quotation text and should be treated as a judgment. Removing a link also removes the heading edge that carries it, which accounts for 34 edge rows. Together the split takes the verified network from 167 to 171 codes and from 1189 to 1201 edges.

A.6 Checks and open items

The build's validators check the network's structure, its round trips and its attributes, but not the content of each edge. The block check places 1301 of the 1326 Index B edge rows (before the 34-row removal) in their printed block, and 25 in another block. The page-image review covers 23 of those 25. Several items remain open. The year of K&K's footnote 5, which Boroch gives as Linton 1945a, was read from the page image at 150 dpi, and a higher-resolution check is still needed. The three duplicate columns need recorded decisions in the crosswalk, and the text analysis must be brought into line with them. The three non-English translations need checking against the page images. The Boroch page numbers need checking against the source works. The 29 link assignments are judgments on the text, as described above.

B Supplementary sankey diagrams at the subgroup level

The NMF comparisons in the body classify the definitions against K&K's seven group letters, first by concept profile and then by wording. The figures here repeat both at the finer grain of K&K's own typology, the roman-numeral subgroups the classification prints within groups C (Normative), D (Psychological), and F (Genetic); groups A (Descriptive), B (Historical), E (Structural), and G (Incomplete) number their codes plainly and have no subgroup level. Each ribbon takes the color of its origin node's group, and node size is the number of codes. The structural panel covers the 149 codes of the largest component's 2-core; the text panel, all 171 codes carrying a quotation.

On the left, A (Descriptive), B (Historical), D-I and F-I reach all seven structural factors, and C-I and D-II reach six. On the right, *group and way of life* and *norms and knowledge* are the largest factors at 26 codes each, and *systems and skills* the smallest at 10. The largest cells are A (Descriptive) on *norms and knowledge* (9 of 22 codes), B (Historical) on *social organization* (7 of 22), and F-I on *complex whole* (6 of 17). No subgroup's codes sit on a single factor; the smallest subgroups in the 2-core, C-II, D-III, D-IV and F-IV (two to four codes each), reach between two and three factors.

On the left, A (Descriptive), C-I and D-II reach all seven factors, and B (Historical), D-I, F-I and F-II each reach six. On the right, *material and transmitted behavior* is the largest factor at 43 codes and draws on D (Psychological, 10), B (Historical, 7) and F (Genetic, 7). *Passed down generations* (34 codes) draws most on D (Psychological, 11) and A (Descriptive, 8), and *symbolic process and adjustment* (26 codes) on F (Genetic, 12 codes, 46% of the factor), B (Historical, 5) and D (Psychological, 4). *Order and operation* (21 codes) draws on F (Genetic, 7), D (Psychological, 4), and A (Descriptive), B (Historical) and C (Normative), 3 each. *Species-specific ability* and *mental and holistic* (17 codes each) both draw on F (Genetic, 6) and C (Normative, 5). *Goals and persons* (13 codes) is mostly D (Psychological, 7 codes, 54% of the factor).

For the definition codes no aggregation is needed, since each code carries its own subgroup. The largest cell holds eight codes (A (Descriptive) on *passed down generations*, 8 of 22 codes), and two cells hold seven (F-I on *symbolic process and adjustment*, 7 of 21; B (Historical) on *material and transmitted behavior*, 7 of 23). No subgroup's codes sit on a single factor. The smallest subgroups, C-II (6 codes), D-III (4) and D-IV (2), reach between two and four factors.

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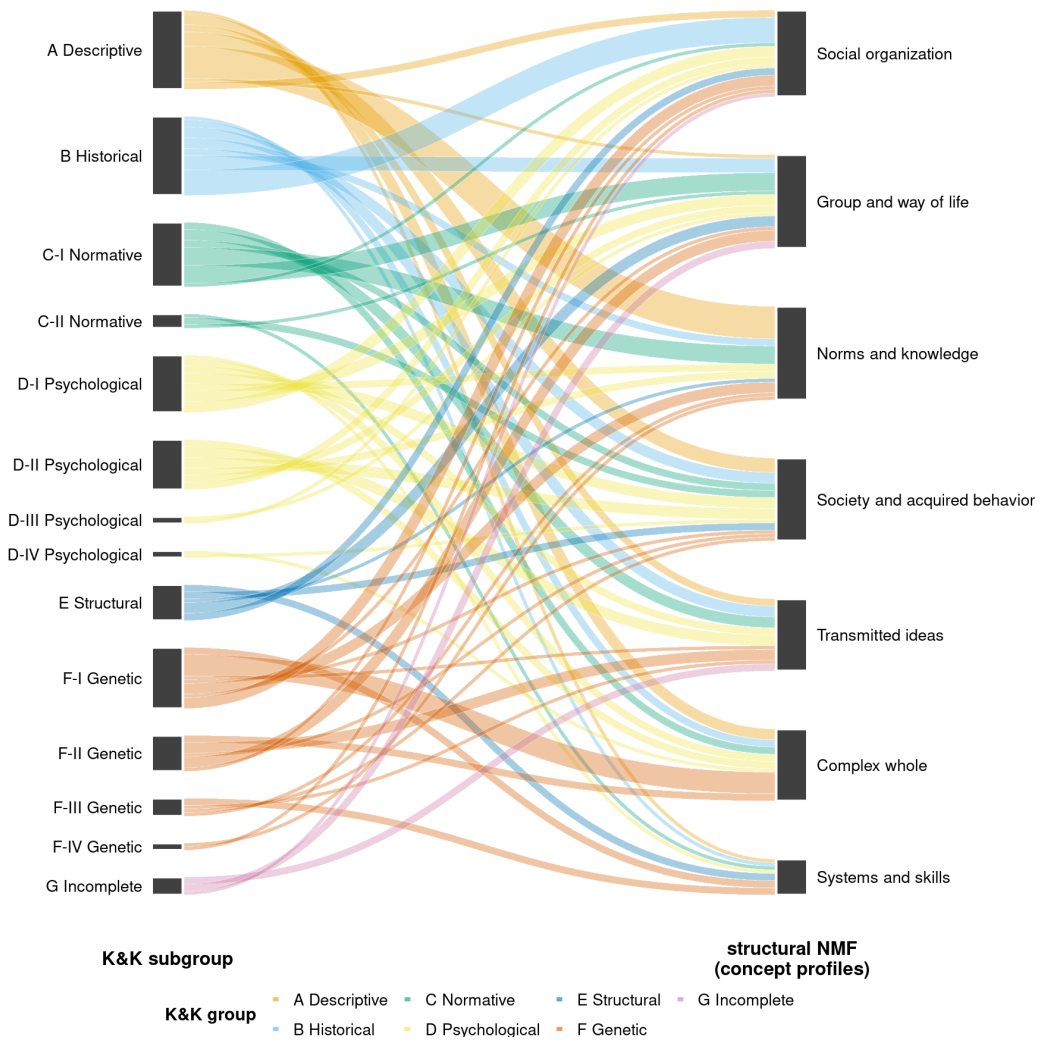


Figure 6: The 149 definition codes of the largest component's 2-core, by their own K&K subgroup (left) and dominant structural-NMF factor (right). Ribbons take the color of their origin node's group.

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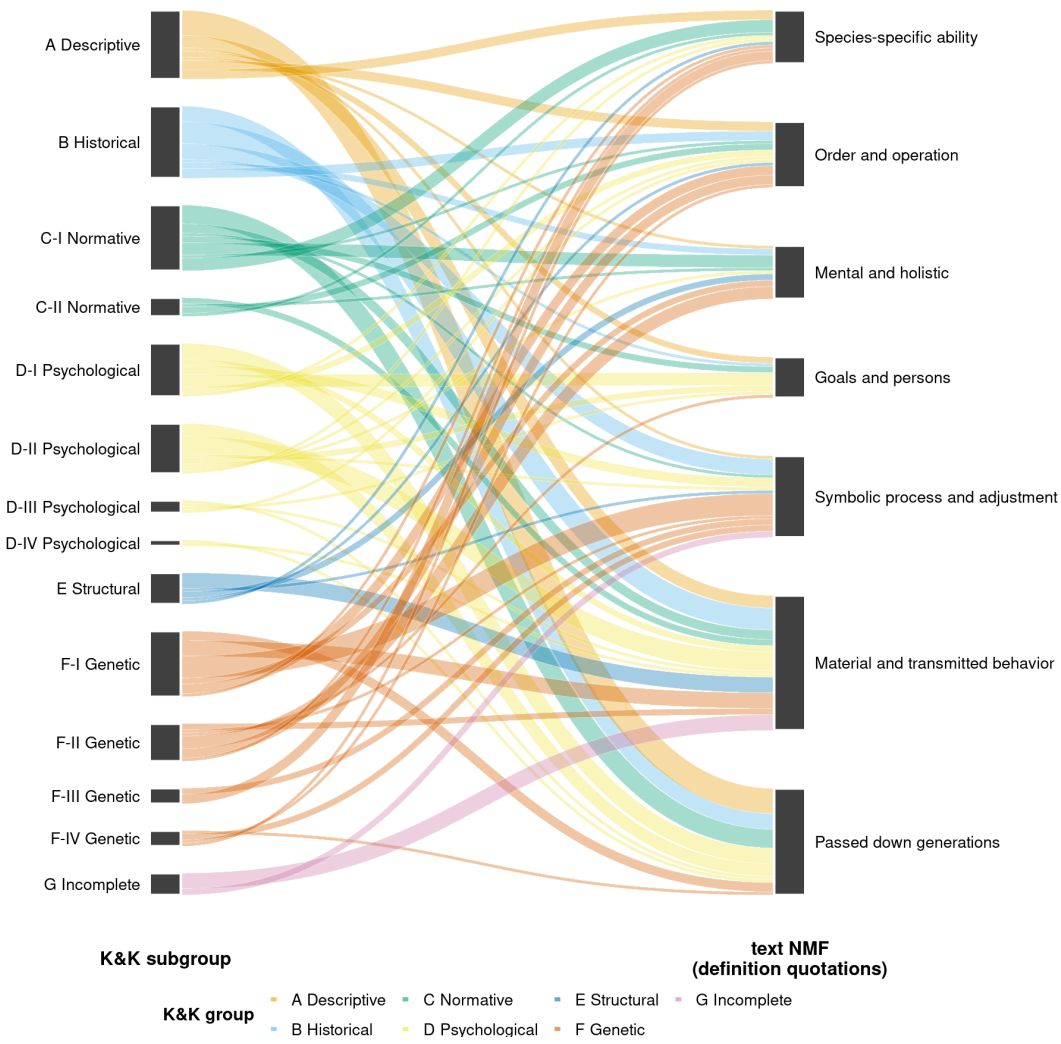


Figure 7: The 171 definition codes carrying a text, by their own K&K subgroup (left) and dominant text NMF factor (right). Ribbons take the color of their origin node's group.

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